Exam Date & Time: 26-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

SEVENTH SEMESTER B.TECH END SEMESTER MAKE-UP EXAMINATIONS, DEC 2024

Remote Sensing and GIS [CIE 4081]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Explain whiskbroom scanning and pushbrom scanning with their respective advantages and disadvantages. (4)

A)

B) Explain normalized density vegetation index (NDVI). (3)

C) Explain what are training sets. Explain how they can be formed using unsupervised classification techniques. (3)

2) Discriminate between the spectral reflectance characteristics of vegetation in visible and infrared regions of the electromagnetic spectrum based on chemical and physical characteristics of their leaves. (4)

A)

B) Explain any three sensors used in Indian satellites for remote sensing purposes. (3)

C) Explain any three applications of remote sensing in the field of hydrology. (3)

3) Explain any two use-cases of thematic maps generated using GIS in the fields of Civil Engineering and real estate. (4)

A)

B) Explain the Electro Magnetic spectrum with a neat sketch. (3)

C) Explain the workings of a confusion matrix and how it is used to assess classification accuracy. (3)

4) Explain the radiation budget equation and explain each of its components with respect to light interacting with an object. (5)

A)

B) Explain how GIS is a better database management tool than scientific visualization sytems, CAD/CAM, and facility management systems (3)

C) Explain parallelopiped classifier. (2)

5) Explain the following terms:

a) banding b) striping c) line drop d) image registration e) random noise. (5)

A)

B) Explain any 3 advantages each of vector and raster models used in GIS. (3)

C) Explain low-pass filtering with an example. (2)

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